Coffee Shop Sales Forecasting & Growth Analysis

Capstone Two: Final Project Report

by Peter Lee

I. Problem Statement

The coffee shop aims to develop its budget for 2026 and evaluate whether an increase is feasible. Specifically, the business would like to determine if a 2–4% increase in sales by the end of 2025 and throughout 2026 is achievable. To support this decision, the company has requested a data-driven analysis to estimate future sales and assess whether this growth target is realistic under current business conditions. The results of this analysis will help guide budgeting decisions and inform potential strategies for revenue growth.

II. Data Description

The dataset used for this analysis was obtained from Kaggle and contains detailed records of coffee sales transactions from March 2024 to March 2025. It includes information on product type, time of purchase, and other temporal features that provide insight into customer behavior and sales patterns.

The target variable for this project is the money column, which represents the revenue generated from each transaction.

To address the business objective, a full data science workflow was implemented, including:

- Data wrangling and cleaning
- Exploratory data analysis (EDA)
- Feature engineering and preprocessing
- Model training and evaluation

A regression model was ultimately developed to predict transaction-level revenue, which is then used to assess overall sales performance and evaluate the feasibility of achieving the desired growth target.

III. Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the distribution of sales, identify key revenue drivers, and uncover patterns that could inform both modeling and business decisions.

A. Distribution of Revenue

The distribution of transaction-level revenue (money) was examined using histograms and boxplots. The results show that most transactions fall within a moderate range, with a smaller number of higher-value purchases contributing to the upper end of the distribution. (Figure 1)

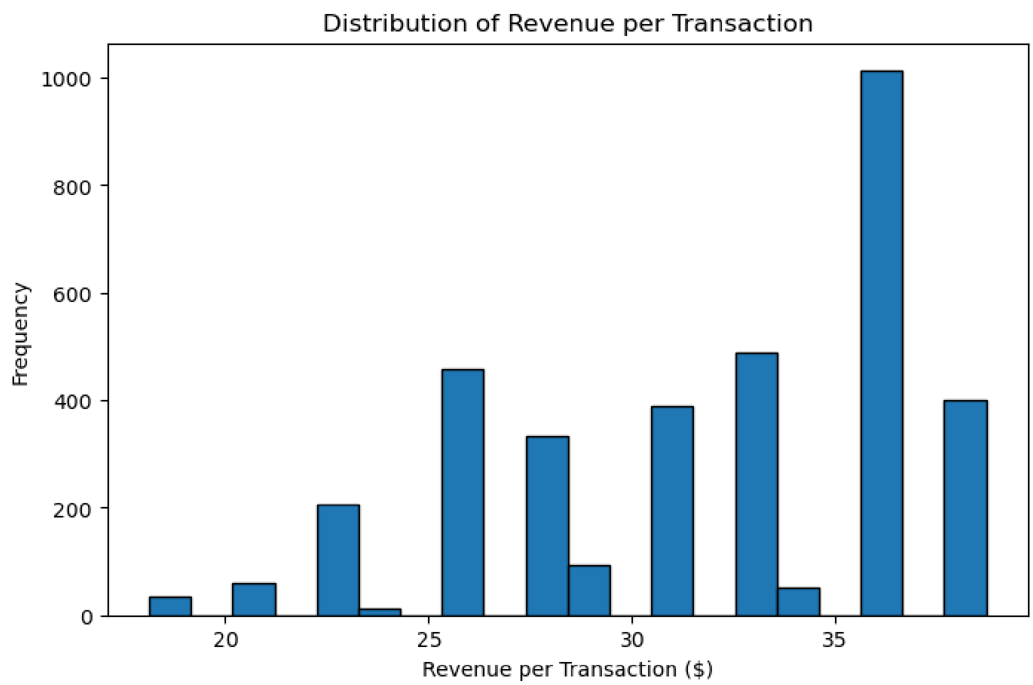

Figure 1. Distribution of Revenue per Transaction

This indicates that while typical purchases are relatively consistent, higher-value drinks play an important role in driving overall revenue.

B. Time-Based Sales Patterns

Sales were analyzed across multiple time-related features, including hour of day, time-of-day categories, weekdays, and months.

- Hourly Trends: Revenue tends to be higher during morning to early afternoon hours, indicating strong demand during peak coffee consumption periods. (Figure 2)

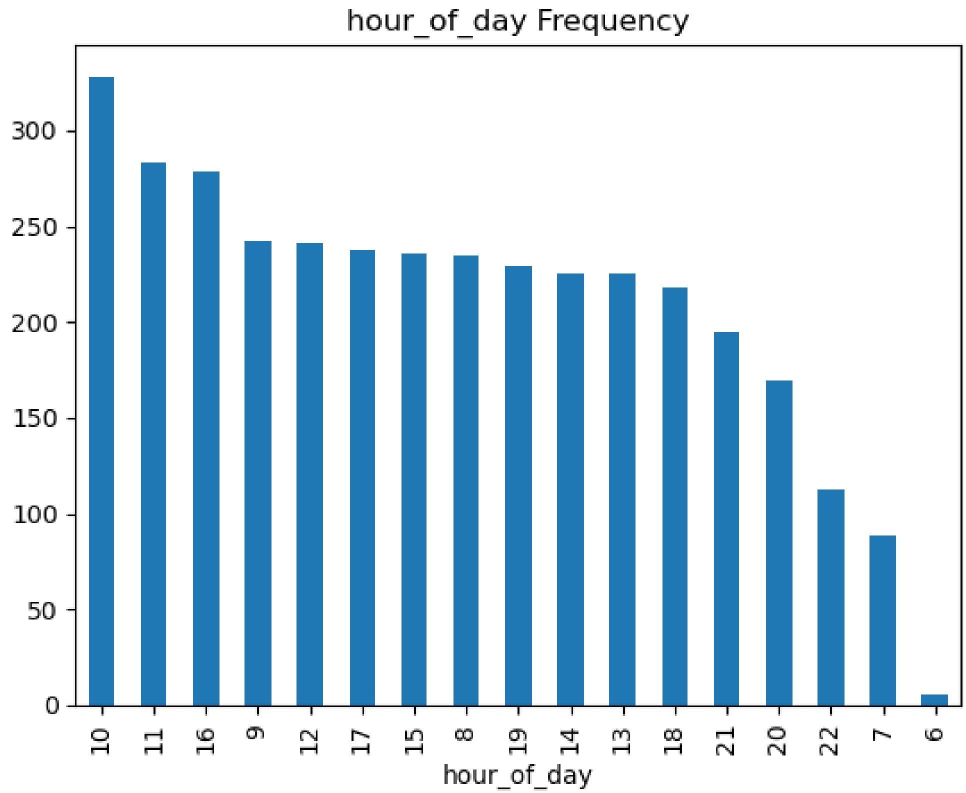

Figure 2. Sales by Hour of Day

- Time of Day: Differences across time-of-day categories further support the importance of temporal patterns.
- Day of Week: Sales vary slightly across weekdays, though differences are relatively small.
- Monthly Trends: Variation in total monthly revenue suggests the presence of seasonal effects, which may impact overall sales performance. (Figure 3)

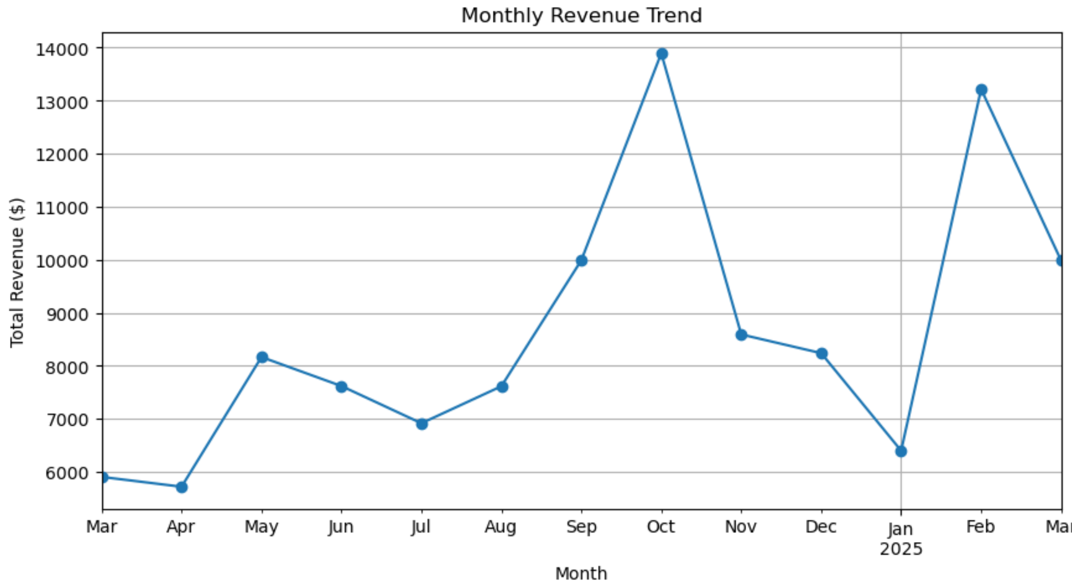

Figure 3. Monthly Revenue Trend

These findings indicate that time-related features contribute to revenue patterns, though their impact appears smaller than other factors.

C. Product-Level Sales Pattern

Analysis of coffee product types revealed significant differences in revenue contribution.

- Certain drinks, such as lattes, americano with milk and cappuccinos, appear more frequently and contribute more to overall revenue (Figure 4)
- Other drinks, such as espresso, generate lower revenue per transaction (Figure 4)

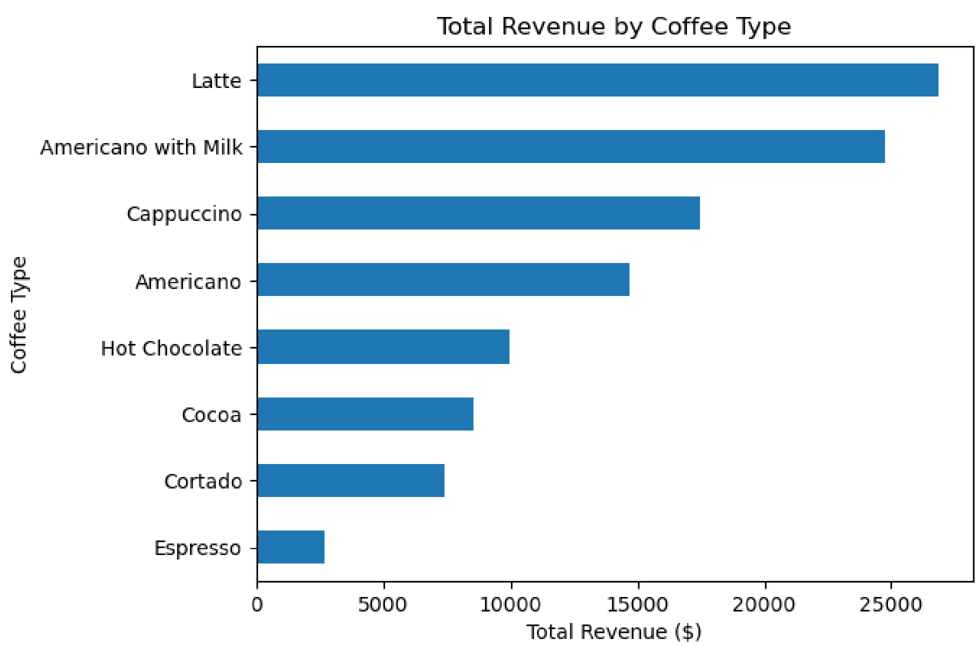

Figure 4. Total Revenue by Coffee Type

This suggests that product type is a key driver of revenue, and that promoting higher-value items could meaningfully increase total sales.

D. Summary of EDA findings

The EDA highlights several important findings:

- Revenue per transaction is concentrated within a moderate range, with higher-value drinks contributing disproportionately to total sales
- Product type is the strongest driver of revenue differences
- Time of day influences sales patterns, particularly during peak hours
- Seasonal variation exists and may impact monthly revenue

These observations suggest that revenue is influenced by structured and predictable factors, supporting the use of regression-based modeling.

IV. Preprocessing & Modeling

A. Data Preprocessing

Before training the models, the dataset was prepared to ensure it was suitable for machine learning algorithms. Categorical variables such as coffee type, time of day, weekday, and month were transformed using one-hot encoding to convert them into a numerical format. The dataset was then split into training and testing sets to evaluate model performance on unseen data. A cross-validation approach was also implemented to ensure that the results were robust and not dependent on a single train-test split.

B. Model Evaluation and Selection

Multiple regression models were evaluated to identify the most effective approach for predicting transaction-level revenue. These included: Linear Regression, Ridge Regression and Random Forest. Model performance was evaluated using R^2 (coefficient of determination) and Mean CV RMSE.

The results showed that:

- Linear Regression and Ridge Regression (with a small alpha value) achieved nearly identical performance
- Both Linear Regression and Ridge Regression models produced high R^2 values and low Mean CV RMSE, indicating strong predictive accuracy (Figure 5)
- Random Forest performed worse than the linear models

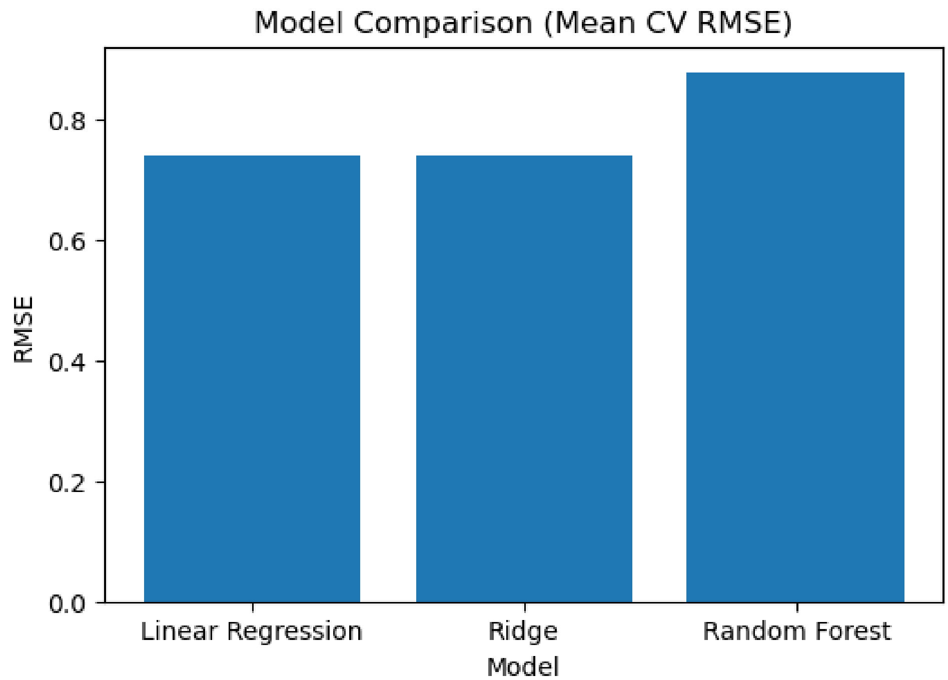

Figure 5. Model Comparison

These results suggest that the relationships in the data are largely linear, and that additional model complexity does not improve performance.

Based on the evaluation results, Linear Regression was selected as the final model. Although Ridge Regression produced similar results, the lack of improvement indicates that regularization was not necessary. Linear Regression was preferred due to its simplicity, interpretability, and ability to clearly explain the relationship between features and revenue.

V. Conclusion and Recommendations

A. Conclusion

Based on the analysis of historical sales data and model performance, current revenue patterns are stable and predictable. The Linear Regression model closely matches observed sales, indicating that the business is operating under consistent and structured demand conditions.

However, the results also show that current trends alone are not sufficient to achieve the desired 2–4% increase in sales. To reach this target, an additional monthly revenue increase of approximately $170–$340 is required.

Encouragingly, this gap is relatively small and can be addressed through targeted, data-driven strategies.

B. Recommendations

1. Promote High-Value Products

   Product type was identified as the strongest driver of revenue, with drinks such as lattes and cappuccinos generating significantly higher revenue per transaction. The business should increase visibility of high-value items, offer bundles or upselling strategies and highlight premium drinks during peak hours.

2. Optimize Peak Sales Periods

   Sales activity is highest during morning to early afternoon hours, indicating key windows of customer demand. The business should ensure adequate staffing during peak hours and run targeted promotions during high-traffic periods. Improving efficiency during these periods can increase total revenue without requiring additional customer volume.

3. Address Seasonal Revenue Fluctuations

   Monthly trends indicate the presence of seasonal dips in revenue, particularly during certain months. The business should introduce seasonal promotions during low-performing periods, adjust product offerings to match seasonal preferences and plan marketing campaigns to stabilize revenue throughout the year. Reducing the impact of low-revenue months can help maintain consistent growth.

4. Opportunities for Further Research
   - Time Series Forecasting: Applying models such as ARIMA or Prophet could provide more precise forecasts of future sales trends.
   - Customer-Level Analysis: Incorporating customer data (if available) could help identify repeat behavior and enable personalized marketing strategies.
   - Price Sensitivity Analysis: Evaluating how changes in pricing affect demand could support more effective pricing strategies.
   - External Factors: Including variables such as weather, holidays, or local events could improve the accuracy of sales predictions.